

Current System Limitations

S.No	Limitation	Impact	Priority
1	Free tier hosting (Vercel)	Limited uptime, app sleeps after inactivity	High
2	Small dataset size	Predictions may lack accuracy for diverse applicants	High
3	Basic security	No advanced encryption or role-based access	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Handles ~100 concurrent users	Migrate to paid cloud hosting (AWS/Render) with auto-scaling
2	Data Storage	CSV dataset stored locally	Move to cloud database (PostgreSQL, MongoDB)
3	Performance	Average response ~1.6s	Optimize ML model, enable caching, use CDN
4	Security	Basic input validation	Add HTTPS, JWT authentication, and encryption

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Expand dataset with real-world credit data	3 months	Improve prediction accuracy
Phase 3	Add user authentication & dashboards	6 months	Secure access and personalized insights
Phase 4	Integrate with banking APIs	9 months	Real-time deployment in financial workflows